Rational Functions & Asymptotes

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.

Factor: $f(x) = \frac{(x-3)(x+3)}{(x-4)(x+3)}$. The factor $(x + 3)$ cancels, giving a hole at $x = -3$. The remaining denominator factor $(x - 4)$ gives a vertical asymptote at $x = 4$.

2 **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the y-coordinate of the hole in the function from the previous problem.

Simplified form (away from the hole): $f(x) = \frac{x-3}{x-4}$. At $x = -3$: $f(-3) = \frac{-3-3}{-3-4} = \frac{-6}{-7} = \frac{6}{7}$. Hole at $\left(-3, \frac{6}{7}\right)$.

3 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.

Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{3}{5}$.

4 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$.

5 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

The numerator's degree (2) is exactly one more than the denominator's degree (1), producing a slant asymptote. Dividing: $x^2 + 1 = (x - 2)(x + 2) + 5$, so $h(x) = x + 2 + \frac{5}{x-2}$. As $x \to \pm\infty$, the remainder term vanishes, leaving the slant asymptote $y = x + 2$.

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6 [Foundational] • Target: Visualizing a rational function's asymptotes

The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.

a Identify the vertical asymptote shown. Vertical asymptote at $x = 2$, where the two branches shoot off toward $\pm\infty$.

b Identify the horizontal asymptote shown. Horizontal asymptote at $y = 1$, which both branches approach far from $x = 2$ (this is the reciprocal function $\frac{1}{x}$ shifted right 2 and up 1).

(a) Vertical asymptote at $x = 2$, where the two branches shoot off toward $\pm\infty$.; (b) Horizontal asymptote at $y = 1$, which both branches approach far from $x = 2$ (this is the reciprocal function $\frac{1}{x}$ shifted right 2 and up 1).

7 [Foundational] • Target: End behavior of rational functions

Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \to \pm\infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).

Since the degree of the numerator (3) exceeds the denominator (2) by exactly 1, f has a slant asymptote and $f(x) \to \pm\infty$ as $x \to \pm\infty$, following the line found by dividing $2x^3 - x$ by $x^2 + 1$, approximately $y = 2x$ for large $|x|$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and intercepts

Find the domain, x-intercept(s), and y-intercept of $f(x) = \frac{x-5}{x^2-4}$.

Domain: all reals except $x = \pm 2$ (where the denominator is zero), so $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$. x-intercept: set numerator to 0: $x = 5$, giving $(5, 0)$. y-intercept: $f(0) = \frac{-5}{-4} = \frac{5}{4}$, giving $\left(0, \frac{5}{4}\right)$.

9 [Intermediate] • Target: Solving rational equations and inequalities

Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.

$x + 1 = 2(x - 3) = 2x - 6$, so $-x = -7$, $x = 7$. Check: $x - 3 = 4 \neq 0$, valid. Solution: $x = 7$.

10 [Intermediate] • Target: Solving rational equations and inequalities

Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.

Critical points: $x = 1$ (zero) and $x = -2$ (excluded). Testing intervals: for $x < -2$, both factors negative, ratio positive. For $-2 < x < 1$, numerator negative, denominator positive, ratio negative. For $x > 1$, both positive, ratio positive. Solution: $x < -2$ or $x > 1$, i.e. $(-\infty, -2) \cup (1, \infty)$.

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11 **[Intermediate]** • **Target: Visualizing a function with a slant asymptote**

The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.

a Find the vertical asymptote and confirm it matches the graph. Vertical asymptote at $x = 1$, matching the break in the graph shown there.

b Use polynomial division to find the slant asymptote equation. Dividing: $x^2 = (x-1)(x+1) + 1$, so $f(x) = x + 1 + \frac{1}{x-1}$. Slant asymptote: $y = x + 1$, matching the line the branches hug for large $|x|$.

c Explain why this function has no horizontal asymptote. Since the numerator's degree (2) exceeds the denominator's degree (1), the function grows without bound rather than leveling off, so there is no horizontal asymptote — only the slant asymptote found above.

(a) Vertical asymptote at $x = 1$, matching the break in the graph shown there.; (b) Dividing: $x^2 = (x-1)(x+1) + 1$, so $f(x) = x + 1 + \frac{1}{x-1}$. Slant asymptote: $y = x + 1$, matching the line the branches hug for large $|x|$.; (c) Since the numerator's degree (2) exceeds the denominator's degree (1), the function grows without bound rather than leveling off, so there is no horizontal asymptote — only the slant asymptote found above.

12 **[Intermediate]** • **Target: Building a rational function from given features**

Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

Start with $\frac{2(x+1)}{x+1} \cdot \frac{1}{x-3}$-style construction: use $f(x) = \frac{2(x+1)^2}{(x+1)(x-3)}$. The common factor $(x+1)$ creates the hole at $x = -1$; the remaining $(x-3)$ in the denominator gives the vertical asymptote; matching leading coefficients (2 over 1) gives horizontal asymptote $y = 2$.

13 **[Intermediate]** • **Target: Building a rational function from given features**

Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

A horizontal asymptote occurs when the numerator's degree is less than or equal to the denominator's degree, while a slant asymptote requires the numerator's degree to be exactly one more than the denominator's. These are mutually exclusive degree conditions, so a rational function has at most one of the two (or neither, if the numerator's degree exceeds the denominator's by more than 1).

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14 [Intermediate] • Target: Rational functions in context

The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \frac{50t}{t^2+4}$ (mg/L). Find the horizontal asymptote and interpret it in context.

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$. In context, this means the drug concentration approaches 0 mg/L as time goes on — the drug is eventually fully eliminated from the bloodstream.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 4.01 Inverse & Composite Functions

Level 3 · AdvancedLet $R(x) = \frac{2x^2-8}{x^2-5x+6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

(a) Domain: Factor the denominator: $x^2-5x+6 = (x-2)(x-3)$. Thus the domain is all real numbers except $x \neq 2$ and $x \neq 3$, or $(-\infty, 2) \cup (2, 3) \cup (3, \infty)$. (b) Holes: Factor the numerator: $2(x^2-4) = 2(x-2)(x+2)$. The factor $(x-2)$ cancels: $R(x) = \frac{2(x+2)}{x-3} \quad (x \neq 2)$. Evaluating at $x = 2$: $y = \frac{2(2+2)}{2-3} = \frac{8}{-1} = -8$. Hole coordinates: $(2, -8)$. (c) Asymptotes: Vertical asymptote at the remaining denominator zero: $x = 3$. Degrees of nu.

16 [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2-4 = (x-2)(x+2)$. The numerator $3x^2+1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3+1/x^2}{1-4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

17 [Challenging] • Target: Building a rational function from given features • Cross-Lesson: 4.03 Parametric Equations & Modeling

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2+5x-3}{x^2+4x+3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x+3)$ cancels for $x \neq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x+1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x-1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

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18 [Challenging] • Target: Rational functions in context • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

The rate of change switches from decreasing to increasing at the inflection point. Taking the second derivative (or analyzing symmetric rates of change): $f'(x) = 3x^2 - 6x - 9, \quad f''(x) = 6x - 6 = 0 \Rightarrow x = 1.$ For x . For $x > 1$, $f''(x) > 0$, so f is concave up on $(1, \infty)$. The inflection point is $(1, f(1)) = (1, 1 - 3 - 9 + 2) = \mathbf{(1, -9)}$.

19 [Challenging] • Target: Vertical asymptotes and holes • Cross-Lesson: 1.03 Rates of Change & Concavity

Level 1 · FoundationsConsider the restricted quadratic function $g(x) = 2(x - 3)^2 + 5$ defined on the domain $[3, \infty)$. Explain why g is invertible on this domain, find the formula for $g^{-1}(x)$, and state the domain of g^{-1}.

Invertibility Justification: The vertex of the parabola is at $(3, 5)$. On $[3, \infty)$, $g(x)$ is strictly increasing and therefore one-to-one, satisfying the horizontal line test. Find Inverse Formula: Set $y = 2(x - 3)^2 + 5$ for $x \geq 3$: $x = 2(y - 3)^2 + 5 \Rightarrow x - 5 = 2(y - 3)^2 \Rightarrow (y - 3)^2 = \frac{x-5}{2}$. Taking the positive square root because $y \geq 3$: $y - 3 = \sqrt{\frac{x-5}{2}} \Rightarrow g^{-1}(x) = 3 + \sqrt{\frac{x-5}{2}}$. Domain of g^{-1}: The range of g is.

20 [Challenging] • Target: Horizontal and slant asymptotes • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Level 3 · AdvancedExperimental laboratory data pairs (x, y) are transformed by computing $Y = \ln(y)$. A linear regression performed on (x, Y) yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for y as an exponential function of x in the form $y = ab^x$, giving a and b to three decimal places. Then predict the value of y when $x = 5$.

Step 1 (Exponentiate the Linear Equation): $\ln(y) = -0.42x + 2.85 \Rightarrow y = e^{-0.42x + 2.85} = e^{2.85} \cdot e^{-0.42x} = e^{2.85} \cdot (e^{-0.42})^x.$ Step 2 (Compute Parameters): $a = e^{2.85} \approx 17.28775 \approx 17.288, \quad b = e^{-0.42} \approx 0.65705 \approx 0.657.$ The exponential model is $y = 17.288(0.657)^x$ (or $y = 17.288e^{-0.42x}$). Step 3 (Predict for x = 5): $y(5) = 17.288 \cdot e^{-0.42 \times 5} = 17.288 \cdot e^{-2.1} \approx 17.288 \cdot 0.122456 \approx \mathbf{2.117}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Vertical asymptotes and holes

Find the hole and vertical asymptote of $f(x) = \frac{(x+2)(x-5)}{(x+2)(x-6)}$.

a. Hole at $x = 6$; vertical asymptote at $x = -2$.

b. **Hole at $x = -2$; vertical asymptote at $x = 6$.** ✓

c. Vertical asymptotes at $x = -2$ and $x = 6$, no hole.

d. Hole at $x = 5$; vertical asymptote at $x = 6$.

The factor $(x + 2)$ cancels, giving a hole at $x = -2$. The remaining denominator factor $(x - 6)$ gives a vertical asymptote at $x = 6$.

22 [Foundational] • Target: Horizontal and slant asymptotes

Find the slant asymptote of $h(x) = \frac{x^2+x+7}{x-7}$.

a. $y = x + 9$

b. $y = x + 7$

c. $y = x - 8$

d. $y = x + 8$ ✓

Dividing $x^2 + x + 7$ by $x - 7$: quotient $= x + 8$, with a remainder that vanishes as $x \to \pm\infty$. Slant asymptote: $y = x + 8$.

23 [Foundational] • Target: Visualizing a rational function's asymptotes

Solve the inequality $\frac{x+6}{x-3} > 0$.

a. $(-\infty, 3) \cup (-6, \infty)$

b. $(-\infty, -6) \cup (3, \infty)$ ✓

c. $[-6, 3]$

d. $(-6, 3)$

Critical points: $x = -6$ (zero) and $x = 3$ (excluded). The expression has the same sign as $(x + 6)(x - 3)$, which is positive outside $[-6, 3]$. Solution: $(-\infty, -6) \cup (3, \infty)$.

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24 [Foundational] • Target: End behavior of rational functions

The graph of the function $f(x) = a \cdot 2^{x-h} + k$ is shown below. The graph passes through the points $(2, 9)$ and $(4, 33)$. The horizontal asymptote is $y = 1$. Find the value of a.

a. **4** ✓

b. 2

c. 1/2

d. 8

The function is given by $f(x) = a \cdot 2^{x-h} + k$. The horizontal asymptote is $y = 1$, which means $k = 1$. So, the function can be written as $f(x) = a \cdot 2^{x-h} + 1$. We are given two points on the graph: $(2, 9)$ and $(4, 33)$. Using the point $(2, 9)$: $9 = a \cdot 2^{2-h} + 1$ $8 = a \cdot 2^{2-h}$ (*) Using the point $(4, 33)$: $33 = a \cdot 2^{4-h} + 1$ $32 = a \cdot 2^{4-h}$ (**) Now we have a system of two equations with two unknowns, a and h: 1) $8 = a \cdot 2^{2-h}$ 2).

25 [Foundational] • Target: Domain and intercepts

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. **Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$** ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $x \neq 2$.

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26 [Foundational] • Target: Solving rational equations and inequalities

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

27 [Foundational] • Target: Visualizing a function with a slant asymptote

Find all rational functions with vertical asymptotes at $x = \pm 1$ and horizontal asymptote $y = 0$.

a. $\frac{x}{x^2 - 1}$ only

b. $\frac{1}{x - 1}$ only

c. Any polynomial

d. **Functions of the form $\frac{k}{x^2 - 1}$** ✓

$\frac{k}{(x - 1)(x + 1)}$ for any nonzero k.

28 [Foundational] • Target: Building a rational function from given features

The graph of the rational function $g(x) = \frac{ax^2 + bx + 1}{x^2 - 4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

The horizontal asymptote of a rational function $g(x) = \frac{P(x)}{Q(x)}$ where the degree of $P(x)$ is equal to the degree of $Q(x)$ is the ratio of the leading coefficients. In this case, the leading coefficient of the numerator is a and the leading coefficient of the denominator is 1. Therefore, the horizontal asymptote is $y = \frac{a}{1} = a$. We are given that the horizontal asymptote is $y = 2$, so $a = 2$. The vertical asymptotes occur at the zeros of the denominator that are not zeros of the numerator.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Rational functions in context**

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. **10 ✓**

b. 16

c. -2

d. 2

We determine the constants a, c, and d step-by-step: 1. **Use the x-intercept:** The point $(-4, 0)$ lies on the graph, meaning $f(-4) = 0$. For a rational function, the numerator must equal zero at an x-intercept: $a(-4) + 8 = 0 \Rightarrow -4a = -8 \Rightarrow a = 2$ 2. **Use the horizontal asymptote:** For $f(x) = \frac{ax+8}{cx+d}$, as $x \to \pm\infty$, the horizontal asymptote is given by the ratio of the leading coefficients: $y = \frac{a}{c} = 2$ Since $a = 2$, we have.

30 **[Intermediate]** • **Target: Vertical asymptotes and holes**

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. **a = 5, b = 3 ✓**

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

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31 [Intermediate] • Target: Horizontal and slant asymptotes

Consider the function $f(x) = 2^x$. Which of the following points lies on the graph of $f(x)$?

a. $(1, 1)$

b. $(2, 4)$ ✓

c. $(3, 6)$

d. $(0, 2)$

The function is given by $f(x) = 2^x$. We need to find which of the given points satisfies this equation. For the point $(1, 1)$: $f(1) = 2^1 = 2$. Since $2 eq 1$, this point is not on the graph. For the point $(2, 4)$: $f(2) = 2^2 = 4$. Since $4 = 4$, this point is on the graph. For the point $(3, 6)$: $f(3) = 2^3 = 8$. Since $8 eq 6$, this point is not on the graph. For the point $(0, 2)$: $f(0) = 2^0 = 1$. Since $1 eq 2$, this point is not on the graph. Therefore, the point $(2, 4)$ lies on the graph of.

32 [Intermediate] • Target: Visualizing a rational function's asymptotes

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int: $x = 3, -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3, -2$; VA: $x = \pm 2$

d. **x-int:** $x = 3$**; VA:** $x = 2$ **only** ✓

Num: $(x - 3)(x + 2)$. Den: $(x - 2)(x + 2)$. Cancel $(x + 2)$: hole at $x = -2$. VA at $x = 2$. x-int: set num of simplified function $= 0$: $x = 3$.

33 [Intermediate] • Target: End behavior of rational functions

Which function has a slant asymptote?

a. $f(x) = \frac{1}{x^2}$

b. $f(x) = \frac{x}{x^2 + 1}$

c. $f(x) = \frac{x^2 + 1}{x}$ ✓

d. $f(x) = \frac{x + 1}{x + 2}$

Degree numerator = degree denominator + 1: slant asymptote.

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34 [Intermediate] • Target: Domain and intercepts

Which function has no vertical asymptote?

a. $f(x) = \frac{x}{x-1}$

b. $f(x) = \frac{1}{x^2}$

c. $f(x) = \frac{x^2+1}{x^2+1} = 1$ ✓

d. $f(x) = \frac{1}{x}$

No denominator zero possible when simplified.

35 [Intermediate] • Target: Solving rational equations and inequalities

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

36 [Intermediate] • Target: Visualizing a function with a slant asymptote

Consider the function $f(x) = \log_3(ax + b)$, where a and b are positive constants. If the graph of $y = f(x)$ passes through the points $(1, 2)$ and $(10, 3)$, find the value of a.

a. 3

b. **1** ✓

c. 1/3

d. 9

The function is given by $f(x) = \log_3(ax + b)$. We are given two points that lie on the graph of this function: $(1, 2)$ and $(10, 3)$. Substituting the first point $(1, 2)$ into the function, we get: $2 = \log_3(a(1) + b)$ $2 = \log_3(a + b)$ By the definition of logarithms, this can be rewritten in exponential form as: $3^2 = a + b$ $9 = a + b$ (Equation 1) Substituting the second point $(10, 3)$ into the function, we get: $3 = \log_3(a(10) + b)$ $3 = \log_3(10a + b)$ In exponential form, this is: $3^3 = 10a + b$.

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37 [Intermediate] • Target: Building a rational function from given features

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R?

a. 2 ✓

b. 3

c. $\log_3 9$

d. $\frac{1}{2}\log_3 18$

The function is given by $f(x) = a \cdot b^x$. It passes through $(1, 6)$ and $(3, 54)$. \\ We have two equations: \\ 1) $6 = a \cdot b^1$ \\ 2) x \\ $9 = b^2$ \\ Since b is a positive constant, $b = 3$. \\ Substitute $b = 3$ into equation (1): \\ $6 = a \cdot 3$ \\ $a = 2$ \\ So, the function is $f(x) = 2 \cdot 3^x$. \\ The related function is $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. Substituting $a = 2$ and $b = 3$, we get: \\ $g(x) = \log_3\left(\frac{2 \cdot 3^x}{2}\right)$ \\ $g(x) = \log_3(3^x)$ \\ Us.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rational functions in context • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

The graph of the function $g(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(3, 24)$. If a and b are positive constants, what is the value of a?

a. 1.5

b. 2

c. **3** ✓

d. 6

The function is given by $g(x) = a \cdot b^x$. Since the graph passes through $(1, 6)$ and $(3, 24)$, we can set up a system of equations: 1. $6 = a \cdot b^1$ 2. $24 = a \cdot b^3$ From equation (1), we can express a in terms of b: $a = \frac{6}{b}$. Substitute this expression for a into equation (2): $24 = \left(\frac{6}{b}\right) \cdot b^3$ Simplify the equation: $24 = 6 \cdot b^{3-1}$ $24 = 6 \cdot b^2$ Now, solve for b^2: $b^2 = \frac{24}{6}$ $b^2 = 4$ Since b is a positive constant, we take the po.

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39 [Challenging] • Target: Vertical asymptotes and holes • Cross-Lesson: 1.03 Rates of Change & Concavity

A scientist is modeling the population growth of a rare species of bioluminescent fungi. The population, $P(t)$, in thousands, after t months is given by the function $P(t) = P_0 e^{kt}$, where P_0 is the initial population and k is the growth constant. Observations show that after 2 months, the population has doubled from its initial size. Furthermore, after 6 months, the population is 16,000. What is the initial population P_0?

a. 2,000 ✓

b. 4,000

c. 8,000

d. 1,000

Let the population at time t be $P(t) = P_0 e^{kt}$. From the problem statement, we have: 1. After 2 months, the population has doubled: $P(2) = 2P_0$. $P_0 e^{k \cdot 2} = 2P_0$ $e^{2k} = 2$ Taking the natural logarithm of both sides: $2k = \ln(2)$ $k = \frac{\ln(2)}{2}$ 2. After 6 months, the population is 16,000: $P(6) = 16000$. Using the value of k we found: $P(6) = P_0 e^{\frac{\ln(2)}{2} \cdot 6} = 16000$ $P_0 e^{3\ln(2)} = 16000$ $P_0 e^{\ln(2^3)} = 16000$ $P_0 \cdot 2^3 = 16000$ $P_0 \cdot 8 = 16000$.

40 [Challenging] • Target: Horizontal and slant asymptotes • Cross-Lesson: 2.01 Exponential Functions & Growth Models

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(2, 36)$. If $b > 0$ and $beq1$, find the value of a.

a. 4 ✓

b. 3

c. 12

d. 1

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(2, 36)$. Using the point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the point $(2, 36)$, we have: $36 = a \cdot b^2$ (Equation 2) Divide Equation 2 by Equation 1: $\frac{36}{12} = \frac{a \cdot b^2}{a \cdot b}$ $3 = b$ Now substitute the value of b into Equation 1: $12 = a \cdot 3$ $a = \frac{12}{3}$ $a = 4$ Thus, the value of a is 4.

41 [Challenging] • Target: Visualizing a rational function's asymptotes • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. What is the value of b?

a. 2

b. 3 ✓

c. 4

d. 6

The function is given by $f(x) = a \cdot b^x$. We are given two points on the graph: $(1, 6)$ and $(2, 18)$. Substituting the first point into the function, we get: $6 = a \cdot b^1$ (Equation 1) Substituting the second point into the function, we get: $18 = a \cdot b^2$ (Equation 2) To find b, we can divide Equation 2 by Equation 1: $\frac{18}{6} = \frac{a \cdot b^2}{a \cdot b}$ $3 = b$ Therefore, the value of b is 3.

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42 [Challenging] • Target: End behavior of rational functions • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

The graph of the exponential function $f(x) = a \cdot 2^x$ passes through the point $(3, 40)$. What is the value of a?

a. $\frac{5}{2}$ ✓

b. 5

c. 10

d. 20

The function is given by $f(x) = a \cdot 2^x$. We are given that the graph passes through the point $(3, 40)$. This means that when $x = 3$, $f(x) = 40$. We can substitute these values into the function equation: $40 = a \cdot 2^3$ Calculate 2^3: $2^3 = 2 \cdot 2 \cdot 2 = 8$ So the equation becomes: $40 = a \cdot 8$ To solve for a, divide both sides by 8: $a = \frac{40}{8}$ $a = 5$ Therefore, the value of a is 5.

43 [Challenging] • Target: Domain and intercepts • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

a. 2

b. **3** ✓

c. 4

d. 6

The amplitude A of a sinusoidal function is half the difference between its maximum and minimum values. \\ Given: Maximum value = 5, Minimum value = -1 \\ Amplitude x is 3. \\ The other parameters (period and phase shift) can be used to find B, C, and D, but are not necessary to determine A.

44 [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider the exponential function $f(x) = 4 \cdot 2^x$. What is the value of $f(3)$?

a. 12

b. 16

c. **32** ✓

d. 24

The function is defined as $f(x) = 4 \cdot 2^x$. To find $f(3)$, substitute $x = 3$ into the function: $f(3) = 4 \cdot 2^3$ Calculate 2^3: $2^3 = 2 \times 2 \times 2 = 8$ Now, multiply by 4: $f(3) = 4 \cdot 8 = 32$ Therefore, the value of $f(3)$ is 32.

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45 [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Which function is decreasing on all of $\mathbb{R}$?

a. $y = e^{x}$

b. $y = x^{2}$

c. $y = e^{-x}$ ✓

d. $y = |x|$

Applying Unit 4 · Functions Involving Parameters, Vectors, and Matrices principles, e^{-x} decreases everywhere. Therefore, the verified correct option is $y = e^{-x}$.

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